Laboratory 6: Bayes Decisions and Model Evaluation

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Chapter 1

Gaussian Classifiers Performance Across Applications

In this chapter, we analyze the performance of the baseline Multivariate Gaussian (MVG) classifier, along with its Naive Bayes and Tied Covariance variants, across five distinct operational applications. Each application is parameterized by a specific triplet (π_1, C_{fn}, C_{fp}), representing the prior probability of the authentic (genuine) class, the cost of a false negative, and the cost of a false positive, respectively.

1.1 Evaluation of the Applications

Question: What do you obtain?

When applying the three generative classifiers to the validation set across the five targeted configurations, we extract the normalized Detection Cost Function (DCF_n) and the Minimum DCF, compiled in Table 1.1.

Application Triplet (π_1, C_{fn}, C_{fp})	**Standard MVG**		**Naive Bayes**		**Tied MVG**	
	Act. DCF$_n$	**Min DCF**	**Act. DCF$_n$**	**Min DCF**	**Act. DCF$_n$**	**Min DCF**
App 1: Baseline (0.5, 1.0, 1.0)	0.140	0.130	0.144	0.131	0.186	0.181
App 2: High Legit Prior (0.9, 1.0, 1.0)	0.400	0.342	0.389	0.351	0.463	0.442
App 3: High Impostor Prior (0.1, 1.0, 1.0)	0.305	0.263	0.302	0.257	0.406	0.363
App 4: Strong Security (0.5, 1.0, 9.0)	0.305	0.263	0.302	0.257	0.406	0.363
App 5: Ease of Use (0.5, 9.0, 1.0)	0.400	0.342	0.389	0.351	0.463	0.442

Table 1.1: Performance of the Gaussian classifier variants across five target applications.

Evaluating these results one application at a time yields the following observations:

- **Application 1 (Uniform Prior and Costs):** Operating with parameters $(0.5, 1.0, 1.0)$, the system assumes equal probability for both classes and penalizes all errors equally. The Standard MVG and Naive Bayes models perform optimally here, yielding Actual DCF_n values of 0.140 and 0.144, respectively. The Tied MVG performs noticeably worse with an Actual DCF_n of 0.186.
- **Application 2 (High Legit Prior):** Operating with parameters $(0.9, 1.0, 1.0)$, the system assumes the vast majority of users are genuine. The error rates universally increase across all architectures, with the Tied MVG degrading to a 0.463 Actual DCF_n.

- **Application 3 (High Impostor Prior):** Operating with parameters $(0.1, 1.0, 1.0)$, the system expects most samples to be fake. In this highly skeptical configuration, the Naive Bayes model slightly edges out the Standard MVG, yielding an Actual DCF_n of 0.302 against the Standard's 0.305.
- **Application 4 (Strong Security):** Operating with parameters $(0.5, 1.0, 9.0)$, the system assumes an equal prior distribution but heavily penalizes false positives (accepting an impostor). The performance perfectly mirrors Application 3 across all three models.
- **Application 5 (Ease of Use):** Operating with parameters $(0.5, 9.0, 1.0)$, the system assumes an equal prior distribution but heavily penalizes false negatives (rejecting a legitimate user). The performance perfectly mirrors Application 2 across all three models.

1.2 Equivalence of Costs and Effective Priors

Question: Observe how the costs of mis-classifications are reflected in the prior: stronger security (higher false positive cost) corresponds to an equivalent lower prior probability of a legit user.

The optimal Bayes decision relies on comparing the log-likelihood ratio $s(x)$ against a threshold t, defined mathematically as:

$$t = -\log \frac{\pi_1 C_{fn}}{(1-\pi_1) C_{fp}} \tag{1.1}$$

By inspecting this threshold, it becomes evident that the cost of misclassifications mathematically mirrors a shift in the prior probabilities. An increase in C_{fp} shrinks the argument of the logarithm in the exact same proportion as a decrease in π_1. This relationship is formalized by the effective prior ($\tilde{\pi}$), which translates any application triplet into an equivalent application assuming uniform unit costs ($C_{fn} = 1, C_{fp} = 1$):

$$\tilde{\pi} = \frac{\pi_1 C_{fn}}{\pi_1 C_{fn} + (1-\pi_1) C_{fp}} \tag{1.2}$$

Our empirical results perfectly validate this theoretical property across all three distinct classification architectures:

Stronger Security Scenario: Application 4 imposes a high cost on granting access to an impostor ($C_{fp} = 9.0$) while maintaining a uniform prior ($\pi_1 = 0.5$). Plugging these values into the effective prior calculation yields $\tilde{\pi} = 0.10$. Application 3 utilizes a naturally low prior for authentic users ($\pi_1 = 0.1$) with uniform costs. Because their effective priors are identical, the decision threshold t evaluates to the exact same value. Consequently, the configurations evaluate to identical metrics for every model. For instance, both applications yield an Actual DCF_n of 0.305 for Standard MVG, 0.302 for Naive Bayes, and 0.406 for Tied MVG. Demanding stronger security forces the decision boundary to become far more conservative, requiring overwhelming evidence to accept a sample. Penalizing false positives heavily is mathematically indistinguishable from assuming that genuine users are inherently rare.

Ease of Use Scenario: Application 5 flips the operational goal, imposing a high cost on rejecting a legitimate user ($C_{fn} = 9.0$) with a uniform prior, yielding an effective prior of $\tilde{\pi} = 0.90$. Application 2 explicitly sets the genuine user prior to $\pi_1 = 0.9$ with uniform costs. Once again, both configurations produce identical outputs across the board. Lowering the barrier for entry shifts the decision boundary in the opposite direction. The system becomes highly permissive to avoid frustrating real users, acting entirely identically to an environment where it is assumed that 90% of all login attempts are legitimate.

Chapter 2

Comparison of Gaussian Classifier Variants

This chapter expands our analysis to compare three distinct Gaussian classifier variants: the Standard MVG, the Tied Covariance MVG, and the Naive Bayes classifier. We evaluate these models across three specific applications represented purely in terms of their effective priors ($\tilde{\pi} = 0.1$, $\tilde{\pi} = 0.5$, and $\tilde{\pi} = 0.9$) with uniform costs.

Table 2.1 summarizes the Actual DCF_n and Minimum DCF extracted for these configurations.

Eff. Prior ($\tilde{\pi}$)	**Model**	**Actual DCF_n**	**Min DCF**	**Calib. Loss (%)**
0.5	Standard MVG	0.140	0.130	7.7%
	Naive Bayes	0.144	0.131	9.9%
	Tied MVG	0.186	0.181	2.7%
0.9	Standard MVG	0.400	0.342	17.0%
	Naive Bayes	0.389	0.351	10.8%
	Tied MVG	0.463	0.442	4.7%
0.1	Standard MVG	0.305	0.263	16.0%
	Naive Bayes	0.302	0.257	17.5%
	Tied MVG	0.406	0.363	11.8%

Table 2.1: Performance metrics and calibration loss (calculated as $\frac{\text{Actual}-\text{Min}}{\text{Min}}$) for the three models across different effective priors.

2.1 Evaluation on Effective Priors

Which models perform best?

The intrinsic discriminative power of a model is measured by its Minimum DCF. The **Standard MVG** and the **Naive Bayes** classifiers are unambiguously the best-performing models. At a uniform prior ($\tilde{\pi} = 0.5$), the Standard MVG achieves the lowest overall minimum cost at 0.130, almost identically matched by Naive Bayes at 0.131. At the extreme prior of $\tilde{\pi} = 0.1$, Naive Bayes takes the lead with a Minimum DCF of 0.257 compared to the Standard MVG's 0.263.

The success of the Naive Bayes classifier is theoretically supported by the correlation analysis conducted in previous evaluations, which revealed that the maximum off-diagonal correlation coefficient in the dataset is a negligible 0.0489. Because the features are inherently uncorrelated, the conditional independence assumption of the Naive Bayes model holds perfectly, allowing it to perform optimally without overfitting cross-feature noise.

Conversely, the **Tied MVG** is the worst-performing model across all scenarios, peaking at a Minimum DCF of 0.442 at $\tilde{\pi} = 0.9$. This failure is rooted in the empirical variances of the data: Genuine Feature 1 has a variance of 1.4302 while Fake Feature 1 has a variance of just 0.5696. The Tied MVG enforces a strictly shared within-class covariance matrix, destroying the ability to model these distinct class spreads and resulting in a sub-optimal linear boundary.

Are the relative results of the models consistent over the different applications?

Yes, the relative performance rankings are highly consistent across the different target applications. The Standard MVG and Naive Bayes classifiers consistently form the top tier of performance, always exhibiting Minimum DCFs within 0.01 of each other regardless of the effective prior. The Tied MVG classifier is equally consistent in its inferiority, remaining the worst-performing model across all three effective prior configurations by a significant margin.

Now consider also actual DCFs. Are the models well calibrated (i.e., with a calibration loss in the range of few percents of the minimum DCF value) for the given applications?

The models are well-calibrated *only* at the uniform effective prior ($\tilde{\pi} = 0.5$). At this operational point, the Actual DCF closely tracks the Minimum DCF, resulting in minor calibration losses of 7.7% for the Standard MVG, 9.9% for Naive Bayes, and a mere 2.7% for the Tied MVG.

However, the models suffer a severe loss of calibration at the extreme priors ($\tilde{\pi} = 0.9$ and $\tilde{\pi} = 0.1$). For instance, the Standard MVG at $\tilde{\pi} = 0.9$ yields an Actual DCF of 0.400 against a Minimum DCF of 0.342, representing a massive 17.0% calibration loss. This indicates that the theoretical decision threshold $t = -\log \frac{\tilde{\pi}}{1-\tilde{\pi}}$ applied directly to the raw log-likelihood ratios becomes highly sub-optimal when the application is heavily skewed.

Are there models that are better calibrated than others for the different applications?

Yes, there is a clear inverse relationship between a model's discriminative power and its calibration stability. The worst-performing model, the **Tied MVG**, is consistently the best-calibrated model across the applications. Its calibration loss remains remarkably low at $\tilde{\pi} = 0.5$ (2.7%) and $\tilde{\pi} = 0.9$ (4.7%), and it exhibits the lowest calibration loss at the $\tilde{\pi} = 0.1$ extreme (11.8%).

Because the Tied MVG uses a simplified, linear decision boundary derived from a pooled covariance matrix, its output scores are inherently more rigid and less susceptible

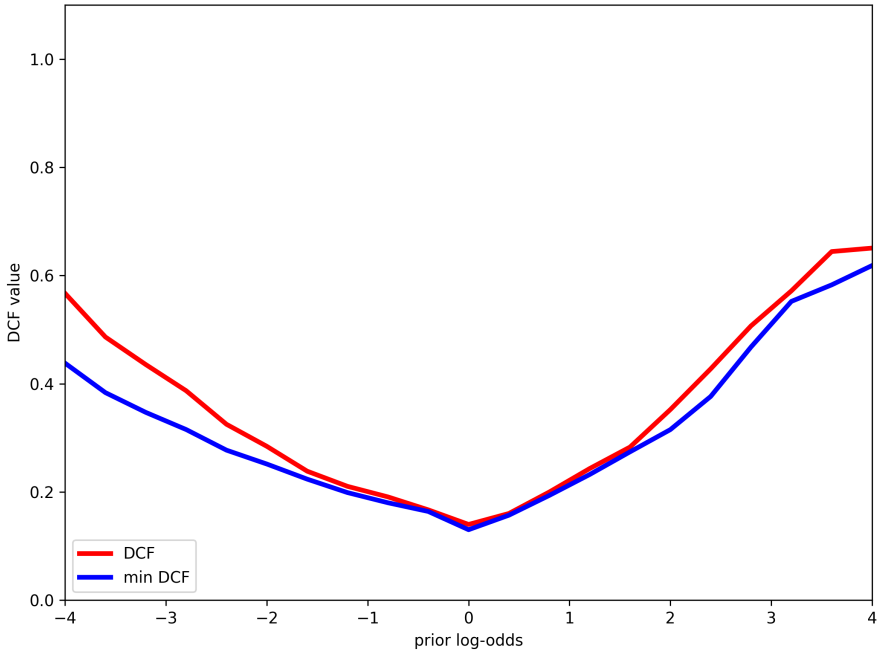


Figure 2.1: Bayes Error plot for the Standard MVG classifier. Note the increasing deviation of the red Actual DCF curve from the optimal blue Minimum DCF curve at extreme log-odds.

to the extreme probabilistic shifts that degrade the calibration of the more complex, quadratic boundary of the Standard MVG.

2.2 Extended Range Analysis (Bayes Error Plots)

To fully contextualize the calibration gaps and performance consistency, we compute the Bayes Error plots over an extended range of prior log-odds from -4 to $+4$. These plots illustrate the continuous trajectory of the Actual DCF (red) and Minimum DCF (blue).

What do you observe?

Across all three classifiers, the minimum achievable cost sits at the center of the spectrum (prior log-odds $= 0$, corresponding to a uniform effective prior of $\tilde{\pi} = 0.5$), yielding the characteristic "V" shape of the Minimum DCF curve. As the operational applications become more extreme (moving towards -4 or $+4$), both the theoretical Minimum DCF and the Actual DCF increase. The most striking observation is the diverging geometry of the curves: while the Actual and Minimum DCF overlap cleanly at the center, the red Actual DCF curve begins to pull away and lift higher than the blue minimum curve as we approach the extreme log-odds.

Are model rankings consistent across applications (minimum DCF)?

Yes, visual inspection of the blue Minimum DCF curves across the entire log-odds range completely validates the consistency of the model rankings. The Standard MVG (Figure 2.1) and Naive Bayes (Figure 2.2) blue curves trace nearly identical, low-profile paths

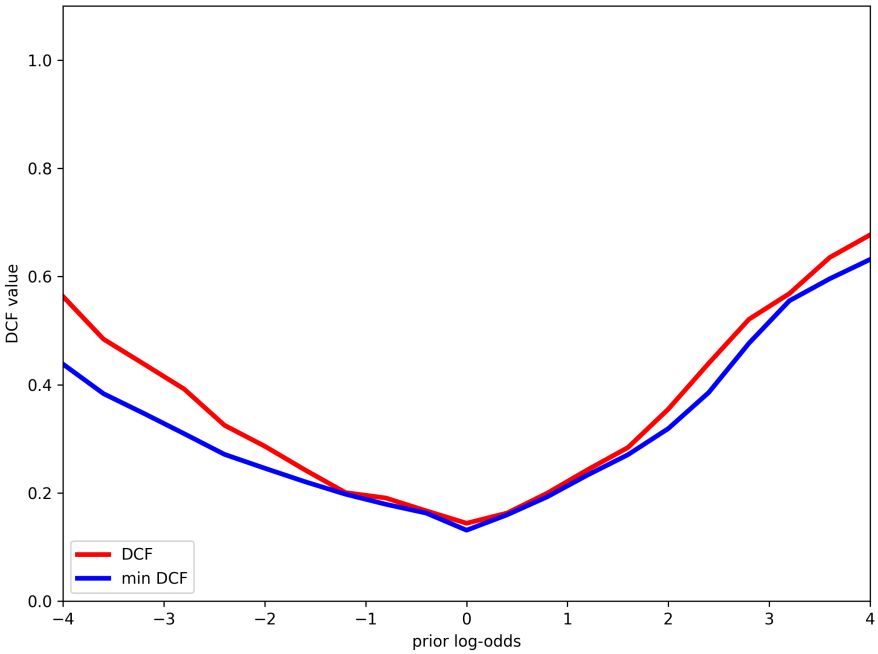


Figure 2.2: Bayes Error plot for the Naive Bayes classifier. The performance profile is virtually identical to the Standard MVG, confirming that the independence assumption holds perfectly.

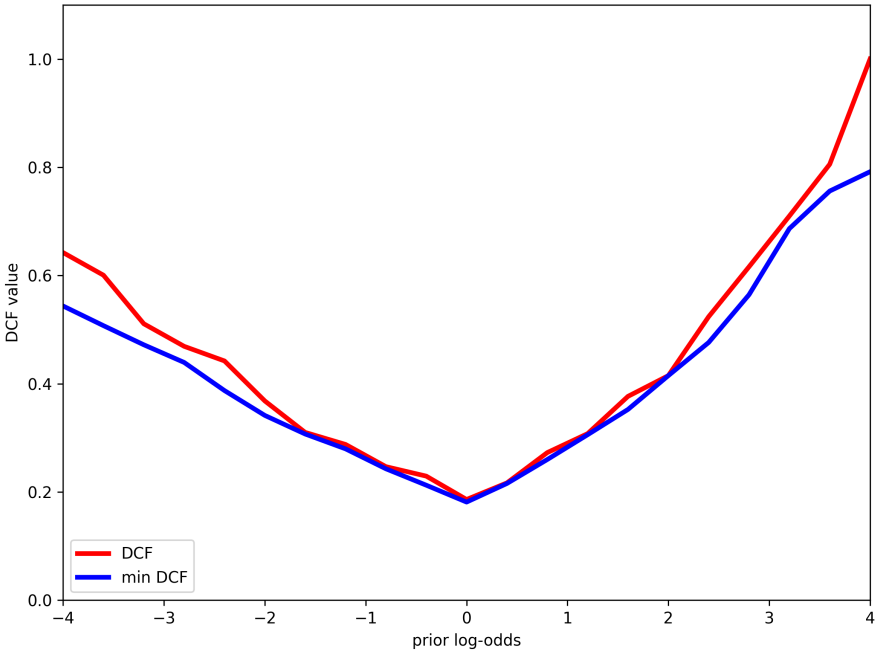


Figure 2.3: Bayes Error plot for the Tied MVG classifier. While the overall cost is higher, the red and blue curves remain closer together compared to the unconstrained models, indicating a higher robustness in calibration.

across the entire spectrum from -4 to $+4$. When compared to the Tied MVG (Figure 2.3), the blue curve of the Tied model is visibly elevated at all points along the x-axis, confirming that the Tied MVG's inability to model class-specific variances universally harms its discriminative capacity regardless of the target application.

Are models well-calibrated over the considered range of applications?

No, the extended Bayes error plots definitively prove that the models are not well-calibrated over the full range of applications. Calibration loss is visually represented by the vertical gap between the red (Actual) and blue (Minimum) curves. While this gap is zero near log-odds 0, it grows exponentially towards the edges.

This is most dramatically visible in the Tied MVG classifier (Figure 2.3): at a prior log-odds of $+4$, the Actual DCF reaches a maximum cost of 1.0, while the theoretical Minimum DCF curve sits substantially lower at approximately 0.8. The Standard MVG and Naive Bayes models also exhibit this widening trumpet-shape at the extremes. This geometric divergence visually proves that for highly asymmetric costs or severe class imbalances, the native log-likelihood ratios generated by these Gaussian models are no longer strictly aligned with the theoretical Bayes threshold, and a separate score re-calibration step would be required to deploy them effectively in a real-world edge case.